

Wajeeh Ul Hassan

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Summary

AI Engineer with hands-on experience in Machine Learning, Generative AI, and MLOps through internships and real-world AI projects. Recently worked as an AI Engineer Intern at CESAI, EME College, building AI-powered inspection solutions. Experienced in developing and deploying machine learning models, RAG applications, and AI systems using Python, FastAPI, LangChain, Docker, and MLflow. Passionate about building practical AI solutions that solve real-world problems.

Work Experience

AI Engineer Intern | CESAI, EME College

Jul 2026 – Sep 2026

- Trained and evaluated a machine learning model on a dataset of approximately 1,000 images for rust and defect detection, achieving 83.5% accuracy.
- Built and integrated an LLM-powered chatbot to provide real-time inspection guidance and assist users during borescope-based vehicle analysis.

AI/ML Engineer Intern | SiberKoza Alpha (NASTP)

July 2025 – Sep 2025

- Developed end-to-end ML pipelines on 80K+ records achieving 88% accuracy, covering ingestion, feature engineering, training, evaluation, and deployment with Docker, MLflow, and DVC.

Achievements

- **Vyrothon 2026 AI Hackathon Winner:** Ranked 1st among 580+ applicants by developing an AI-powered voice agent for real estate lead automation.
- **Algoryc AI Hackathon 2026:** Shortlisted among approximately 40 invite-only participants.
- **Invited Speaker – GDG on Campus COMSATS Abbottabad:** Delivered a webinar on Artificial Intelligence and AI development.

Technical Skills

- **Languages & Tools:** Python, SQL, Git, GitHub Actions, Docker, FastAPI, MLflow, DVC, AWS (EC2, S3)
- **ML & AI:** Machine Learning, Deep Learning, NLP, Generative AI, LLMs, XGBoost, scikit-learn, TensorFlow, PyTorch
- **NLP & Speech AI:** LangChain, LangGraph, RAG, Prompt Engineering, Speech-to-Text, Text-to-Speech Concepts, Conversational AI
- **Data & Databases:** PostgreSQL, MongoDB, ChromaDB, pandas, NumPy, Data Modeling, ETL Pipelines, Great Expectations

Projects

MeetingMind – AI Video & Meeting Assistant

[\[GitHub\]](#)

- Built an end-to-end AI meeting assistant using Python, LangChain, OpenAI APIs, and ChromaDB for multilingual transcription, summarization, action-item extraction, and conversational RAG over video/audio content.
- Implemented document chunking, OpenAI embeddings, vector retrieval, and an interactive Q&A pipeline for context-grounded responses, supporting English, Urdu, and multilingual speech.

Final Year Project – AI-Based Virtual Try-On System

[\[GitHub\]](#)

- Developed an AI-powered virtual try-on system using computer vision, image segmentation, pose estimation, and diffusion models to generate realistic clothing visualization for online shopping applications.
- Designed modular PyTorch + OpenCV pipeline enabling end-to-end inference from raw photo to rendered try-on output.

PurchasePulse – Real-Time Purchase Intent Prediction

[\[GitHub\]](#)

- Trained XGBoost on 66 features achieving 90.23% accuracy / 0.89 ROC-AUC; served via FastAPI with batch and single-inference endpoints, Streamlit drift-detection dashboard, and MLflow tracking.
- Containerized with Docker Compose; DVC-managed data/model versioning ensures fully reproducible workflows.

Education

BSc Artificial Intelligence | NUML, Islamabad

2022 – 2026

- Relevant Coursework: Machine Learning, Deep Learning, Artificial Neural Networks, NLP, Computer Vision

Certifications

- **Supervised Machine Learning: Regression and Classification**
- **Oracle Cloud Infrastructure AI Foundations Associate**
- **Oracle Cloud Infrastructure Generative AI Professional**

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